A Novel Exoskeleton Neuromuscular Interface Based on Motor Unit Action Potential Model Using High-Density sEMG

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***Abstract*— The topical and noninvasive measurement of high-density surface electromyogram (HD-sEMG) signals enables the estimation of human motor unit action potentials (MUAPs) as crucial motor function indicators for human–robot interaction. In this article, we present a 2-D high-density microneedle electrode array using the potassium hydroxide (KOH) bulk etching technique and the flexible printed circuit (FPC). To determine the optimal configuration of the 2-D electrode array, we propose an accurate and efficient neuromuscular analytical model for HD-sEMG that comprehensively analyzes the effects of the microneedle electrode size, interelectrode distance, and location. The experiment was conducted to demonstrate the feasibility and performance of the proposed microneedle-based high-density electrode array for the sEMG-based upper limb exoskeleton's elbow joint angle estimation, specifically in comparison to a commercial wet electrode. The experimental results showed that the proposed microneedle electrode with high spatial resolution was comparable to the wet electrode (Wilcoxon rank-sum test, $p > 0.05$). On average, the correlation and root-mean-squared error (RMSE) of the microneedle electrode array with high space utilization were 7.56% and 19.83% better than those of the wet electrode, respectively. The 2-D high-density microneedle electrode array based on the proposed HD-sEMG model facilitates a novel neural–machine interface for intuitive control of upper limb exoskeletons.**

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***Index Terms*— Electrode configuration, high-density surface electromyogram (HD-sEMG) modeling, high-density, microneedle electrode array, motor unit (MU).**

I. Introduction

SURFACE electromyogram (sEMG) contains the action potentials of motor units (MUs) and has been widely used in many fields, such as medical rehabilitation, competitive sports, and human–robot interaction [1], [2], [3]. High-density sEMG (HD-sEMG) uses multiple closely spaced electrodes overlying a restricted area of the skin to measure human MU action potentials (MUAPs) [4].

The main challenges of high-quality sEMG acquisition are the high impedance of the skin stratum corneum and motion artifacts [5]. The conductive gel in the conventional wet electrode not only significantly reduces the electrode–skin contact impedance but also has good adhesion. Limited by the small electrode area, the HD-sEMG sensors use rigid dry electrodes and have poor performance [6]. The microneedle electrodes have made great progress in materials, structures, and fabrication methods [7]. The introduction of microneedles into HD-sEMG sensors will address these challenges. The microneedles can penetrate the skin stratum corneum and avoid electrode displacement [5]. A majority of studies focus on skin-like, soft, and stretchable properties of sensors [8], [9]; however, few of them have considered the high-density and multichannel properties and electrode configuration (e.g., electrode size, interelectrode distance, and placement). The electrode configuration greatly affects the accuracy of the application [10], [11], [12], [13].

Although a common body of knowledge has been established through publication [14], many of its recommendations are statistical summaries and not applicable to HD-sEMG sensor configuration. The neuromuscular analytical model provides an outlet for interpreting and simulating sEMG signals [15], [16], sEMG filtering [17], and neuropathological diagnosis [18]. Furthermore, it provides a simulation method for electrode configuration [19], such as monopolar and bipolar electrodes. In [20], the effects of electrode size and interelectrode spacing on the detection distance were examined using an approximate method. The work in [21] proposed a 2-D spatial filtering technique without approximation of the integration effect and restriction of electrode shapes. It has

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been noted that monopolar, bipolar, and Laplacian electrode configurations affect the accuracy of force-level classification in simulation and experimental studies [10], [11]. However, with the development of HD-sEMG, the use of a 2-D electrode array composed of monopolar and bipolar montages in HD-sEMG imaging deserves specific attention [22], [23].

The existing analytical models have two challenges to be solved for HD-sEMG. First, most of them consider single-fiber action potentials (SFAPs) recorded by a single electrode [24], [25], or MUAPs [20], [26]. Those models are limited by the complex anatomical structure of the muscle and at high computational costs. However, the computational costs are unaffordable for configurations with tens to hundreds of electrodes [15], as shown in Table S1 in the supplementary material. A few studies discussed approaches to reduce the computation time, such as the notion of the isopotential layer [20] and the principle of reciprocity [27]. The work in [15] reduced the computation time by setting all the computations in a 3-D Fourier domain and using parallel computing. Built on [15], the work in [28] presented a method of macroscale MU electrical source, which was 10% more efficient than parallel computing. However, the simulation efficiency is unsatisfactory when the muscle fiber orientation and curvature are considered or additional MUs are recruited [27]. Moreover, the SFAP and MUAP models neglect the interaction and comprehensive effects of muscle fibers and MUs and therefore cannot be applied to HD-sEMG applications [28]. Second, while they minimally influence the single-channel recordings, specific characteristics have substantial effects on the multichannel results, particularly the MU territory locations in the muscle cross section and the motor end-plate geometry [26].

In this article, we present a 2-D high-density microneedle electrode array with high spatial resolution and space utilization, see Fig. 1. The microneedle electrodes are fabricated from silicon by the potassium hydroxide (KOH) bulk etching technique and integrated with a flexible printed circuit (FPC) board. An accurate and efficient neuromuscular analytical model for HD-sEMG is proposed to simulate the microneedle electrode size, interelectrode distance, and location of the 2-D electrode array for optimal sensor configuration. In our model, we consider a cylindrical muscle, with all the muscle fibers parallel to the axis of the muscle; see Fig. 2. A neighborhood judgment algorithm (NJA) is proposed to uniformly locate the MU territory and muscle fiber in the muscle cross section. The motor end-plates of MU are modeled as a multiband Gaussian distribution. These new features are appropriate for multichannel recordings. Based on the uniformity of muscle fibers and multiband distribution of motor end-plates, a simplified computational method for MUAP generation is proposed to reduce the computational time. The elbow joint angle estimation experiment for the sEMG-based upper limb exoskeleton was performed, and the performance of the microneedle-based high-density electrode array was verified by comparing it with the conventional wet electrode.

The main contributions of the work are twofold. First, a 2-D high-density microneedle electrode array with high spatial resolution and space utilization is designed, which has the promising potential for neural-driven-based applications. Second, we present an accurate and efficient neuromuscular analytical model for HD-sEMG sensor configuration. We optimize the modeling of MU location, muscle fiber location, and motor end-plate distribution for multichannel recordings. The uniformity of the location is in good agreement with the expected data (21.08 ± 1.22 compared with 20.79 fibers/mm^2). The simplified MUAP computational method leads to a sixfold reduction in the computation time of the HD-sEMG, from 1 h in other methods to 10 min in this work.

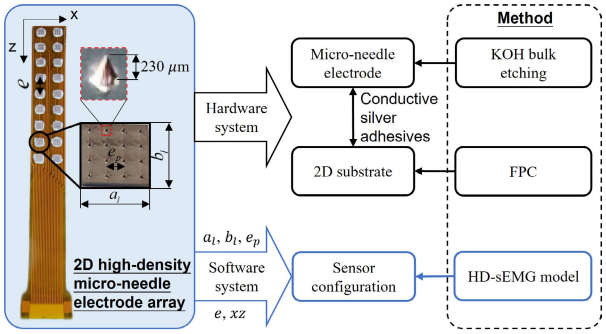

Fig. 1. Schematic of the novel exoskeleton neuromuscular interface.

II. Materials and Methods

A. 2-D High-Density Microneedle Electrode Array

As shown in Fig. 1, we developed a 9 × 2 high-density microneedle electrode array for the sEMG-based upper limb exoskeleton control.

1) Hardware System: The 2-D electrode array consists of two main components: microneedle electrodes and a 2-D flexible substrate. Silicon has good biocompatibility, chemical stability, and high hardness, which can penetrate the skin stratum corneum and achieve long-term sEMG monitoring. The noninvasive silicon microneedle electrode with a height of 230 μm is fabricated by the KOH bulk etching technique. It is then sputtered with a layer of silver with a thickness of 200 nm and finally integrated with the FPC using conductive silver adhesives (Ablebond 84-1LMISR4, Germany).

2) Software System: The 2-D electrode array is characterized by five parameters, including the size $a_l \times b_l$ and the interelectrode distance e of the microneedle electrode, the pin-to-pin distance e_p of the microneedle, and the layout (e.g., 9 × 2) in the xz plane. In addition to its characteristics, the configuration of the sensor also includes the location on the skin and the maximum limit of size. An accurate and efficient neuromuscular analytical model for HD-sEMG (HD-sEMG model in Section II-B) is used to determine the optimal sensor configuration, which is described in Section II-C.

B. HD-sEMG Model

Fig. 2 illustrates the basic modeling setup for HD-sEMG. The Cartesian coordinates are fixed on the muscle center with the z-axis parallel to the muscle fiber direction, and the xy plane is the muscle cross section. The muscle is surrounded by skin and fat tissues whose behaviors are modeled by

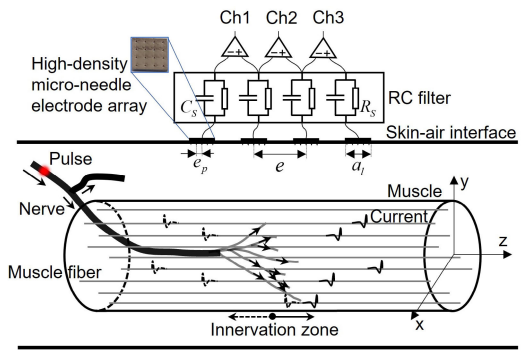

Fig. 2. Illustrative diagram of HD-sEMG model development.

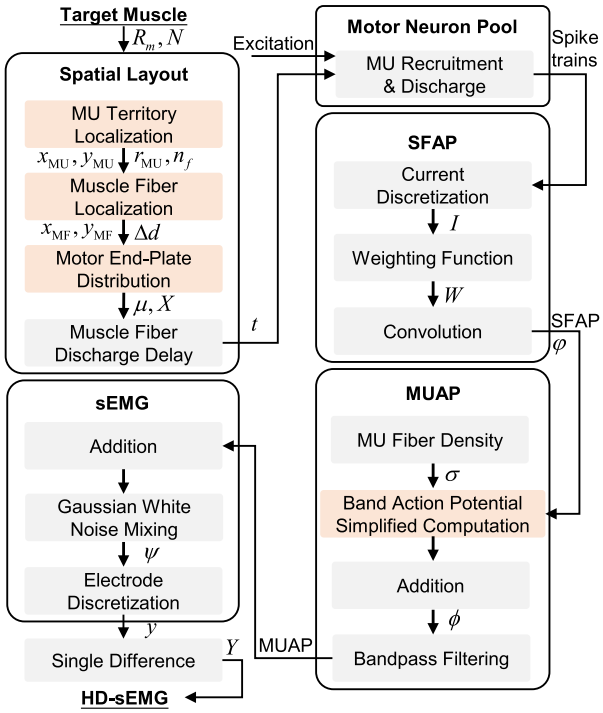

Fig. 3. Flowchart diagram of HD-sEMG model development. x_{MU} and y_{MU} are the x- and y-coordinate of the MU, and x_{MF} and y_{MF} are the x- and y-coordinate of the muscle fiber, respectively.

an RC filter. Pulses from the nerve conduct in both the directions along muscle fibers. The potentials detected by the high-density microneedle electrode array are filtered and amplified.

Fig. 3 shows the flowchart diagram of the HD-sEMG modeling approach. We highlight the main new developments in the figure, that is, the locations of MU territory and muscle fiber, the distribution of motor end-plate, and the simplified computation of MUAP generation. The modeling approach comprises five modules: spatial layout, motor neuron pool, SFAP, MUAP, and sEMG. The spatial layout module establishes an equivalent virtual muscle model and describes the modeling methods of MUs, muscle fibers, and motor end-plates for multichannel recordings. The motor neuron pool module defines how each MU is recruited and the relationship between the excitation and discharge after recruitment. The remaining three modules describe the modeling methods of SFAP, MUAP, and sEMG after the muscle established in

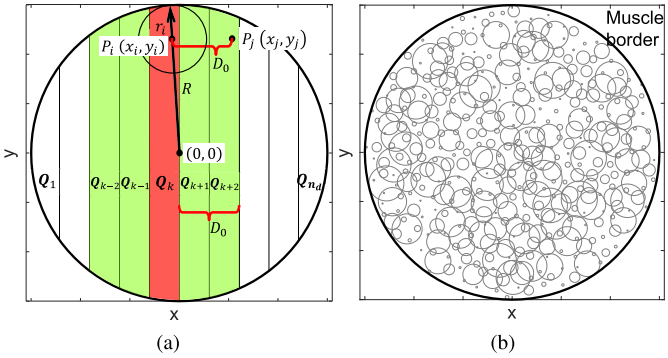

Fig. 4. (a) Illustrative schematic of the NJA algorithm. The red region represents the set where the newly generated point is located, and the green regions represent the sets that need to be calculated. (b) Distribution of $N = 443$ MUs across the muscle cross section. The circles within the muscle border are the MU boundaries.

the first module is activated by the second module. In the following, we describe these modules and mainly highlight the newly developed components.

1) Uniform Distribution of MU Territory Centers: The MU territory centers are evenly covered throughout the muscle cross section based on the experimental study of [29]. We propose the NJA algorithm that makes the position of the MU territory center quasi-random, that is, not limited by a fixed layout and the initial point position. The goal of the NJA algorithm is to generate uniformly distributed MU territory centers at a low time cost.

We assume that the muscle cross section located at (x_c, y_c) is a circular shape with a known radius R_m, and the total number of MUs is N. The desired distance d_0 between two adjacent points in the grid layout is calculated by estimating the number of points distributed over the radius with formula fitting. The NJA algorithm is to calculate the Euclidean distance between the newly generated point and the existing points in turn. If the distance is greater than or equal to d_0, the point is retained; otherwise, it is discarded. Then the next new point is generated randomly until the number of retained points reaches N. The optimization of this algorithm is to reduce the calculation when experiencing a number of points. In fact, only points close to this new point will be calculated.

As shown in Fig. 4(a), starting from the origin $(0, 0)$, the entire muscle cross section is divided into n_d sets by vertical lines that are spaced by $D_0/2$ apart, where $D_0 = d_0/\epsilon$ and constant $\epsilon > 1$ is chosen to compromise the computational cost and distribution uniformity. The sets are denoted as $\boldsymbol{Q}_l$, $l = 1, \ldots, n_d$. As an example, a given point P_i falls in $\boldsymbol{Q}_k$, and it just needs to calculate the Euclidean distance between it and the existing points in $\boldsymbol{Q}_k$, $\boldsymbol{Q}_{k-1}$, $\boldsymbol{Q}_{k-2}$, $\boldsymbol{Q}_{k+1}$, and $\boldsymbol{Q}_{k+2}$. The MU is usually defined as a circular territory with a certain radius [20], [28]. Therefore, the NJA algorithm must ensure that the MU territory center and the entire territory are located within the muscle cross section. Fig. 4(b) shows an example of the distribution of MUs across the muscle cross section by the NJA algorithm. The circles within the muscle border are the MU boundaries. Algorithm 1 further illustrates the outline of the NJA algorithm for uniformly distributed points.

The distribution of muscle fibers in an MU is also confirmed to be uniform [30]. We denote the generated MU_i territory

Algorithm 1 NJA Algorithm

Input : Center (x_c, y_c), R, N, radii $\boldsymbol{r} = \{r_i\}$
Output: $(\boldsymbol{x}, \boldsymbol{y})$
$n = \frac{1+\sqrt{N-4}}{2}$, $d_0 = \frac{R}{\lceil n \rceil}$, and $D_0 = d_0/\epsilon$, $i \leftarrow 1$;
Partition cross section into n_d regions with $\boldsymbol{Q}_l = \varnothing$, $l = 1, \ldots, n_d$;
while $i \leq N$ **do**
  Randomly generate $P_i = (x_i, y_i)$, $x_i, y_i \in [-R, R]$;
  if $|P_i| \leq R - r_i$ **then**
    $k = \left\lfloor \frac{x_i}{D_0/2} \right\rfloor + \left\lceil \frac{R}{D_0/2} \right\rceil + 1$, $\boldsymbol{\psi} = \varnothing$;
    $\boldsymbol{\psi} \leftarrow \boldsymbol{Q}_{k-2} \cup \boldsymbol{Q}_{k-1} \cup \boldsymbol{Q}_k \cup \boldsymbol{Q}_{k+1} \cup \boldsymbol{Q}_{k+2}$;
    if $\forall P_j \in \boldsymbol{\psi}, |P_i P_j| \geq D_0$ **then**
      $\boldsymbol{Q}_k \leftarrow \boldsymbol{Q}_k \cup \{P_i\}$;
      $(\boldsymbol{x}, \boldsymbol{y}) \leftarrow (\boldsymbol{x}, \boldsymbol{y}) \cup \{(x_i + x_c, y_i + y_c)\}$;
      $i \leftarrow i + 1$;

center as $(x^i_{\mathrm{MU}}, y^i_{\mathrm{MU}})$, the radius as r^i_{MU}, the total number of muscle fibers as n^i_f, and the muscle fiber diameter as d^i_f. The uniform location layout can also be obtained by the NJA algorithm.

2) Multiband Gaussian Distribution of Motor End-Plates: The physiological structure on the muscle fiber where the excitation occurs is called the motor end-plate. The motor end-plate position (the z-axis coordinate) varies in different bands of the MU territory. The band division of each MU territory in the muscle cross section is based on two principles: (1) *Proximity principle*: Within the same MU, adjacent muscle fibers have similar properties such as length and motor end-plate position. Based on this, we divide the muscle fibers in the neighborhood into a band. (2) *Simplified calculation principle*: The band division needs the reference and proximity distance. Since the action potential is highly related to the distance between the electrode and the muscle fiber, we take the position of the electrode ($x = 0$) as the reference and the safe distance (e.g., 0.5 mm) that does not affect action potentials as the width of each band to simplify the calculation of action potentials [20].

Fig. 5 illustrates the motor end-plate band division for a given MU_i, $i \in \mathbb{N}$. The reference point is at location $x = 0$. Centered at the reference point, a set of equally distanced arc-shaped dotted lines are crossing MU_i, and a band is formed between two adjacent dotted lines with bandwidth Δd. The average motor end-plate position μ_i for MU_i is assumed to be Gaussian-distributed with a mean of 0 and a standard deviation of σ_{MU}. The motor end-plate position X^j_i of the jth band in MU_i also satisfies the Gaussian distribution and is expressed as

$$X^j_i \sim \mathcal{N}\left(\mu_i, \sigma_b^2\right), \ -L_0/2 \leq X^j_i \leq L_0/2 \tag{1}$$

where σ_b is the standard deviation, $\sigma_b < \sigma_{\mathrm{MU}}$, $j = 1, \ldots, n_b$, n_b is the number of bands crossing MU_i, and L_0 is the range of motor end-plate position.

3) Discharge Delay of Muscle Fibers: Due to the differences in the conduction path and conduction velocity among muscle fibers, the same excitation arrives at motor end-plates

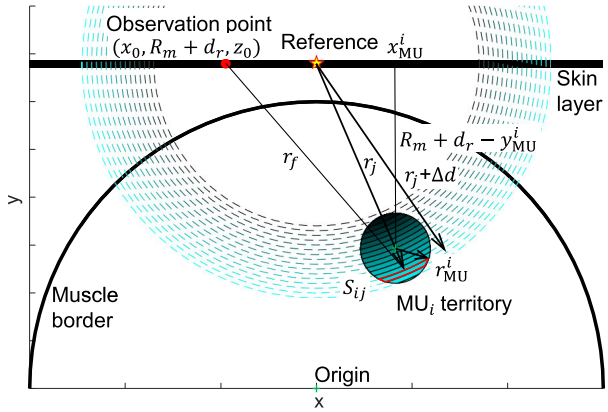

Fig. 5. Illustrative schematic of the motor end-plate band division on the muscle cross section. Centered at the reference point, each band is filled with the same color as indicated by the arc-shaped dotted line. The darker the color, the greater the contribution of action potentials generated by the current source within the band.

at different times, resulting in delayed discharges of muscle fibers. The muscle fiber discharge delay t_{ij} at the jth band in MU_i can be expressed as

$$t_{ij} = t_i + t^j_{bi}, \tag{2}$$

where t_i and t^j_{bi} represent the delays of MU_i among MUs and the jth band among bands in MU_i, respectively, and are calculated as

$$t_i = \frac{\mu_i - \min(\mu_i)}{v_f}, \ t^j_{bi} = \frac{|X^j_i - \mu_i|}{v_p}, \tag{3}$$

and v_f and v_p are the conduction velocities of axons and telodendrias of motor neurons, respectively.

When the excitation in the trigger zone reaches the recruitment threshold of an MU, the motor neuron of this MU starts to fire at a minimum firing rate (MFR). Due to the relationship between stimulus rate and isometric force for MUs, the peak firing rate (PFR) for MU_i is calculated as $\mathrm{PFR}_i = 1.5/T_i$, where T_i is the contraction time of MU_i. The recruitment threshold excitation (RTE) and the contraction time of MU_i are distributed according to the principle in [31]. The linear gain g_e of the excitation–firing rate relationship is calculated as $g_e = (\mathrm{PFR}_N - \mathrm{MFR}/100 - \mathrm{RTE}_N)$. The spike trains of MU_i are obtained with a coefficient of variation of 0.2 in the interspike intervals and t_{ij} added to them.

4) SFAP Modeling: Under the premise of a time-shift-invariant system, SFAP φ_{ij} at the jth band in MU_i can be expressed as the discrete convolution of the current source I and the weighting function W of the volume conductor system [27], [32], [33], namely,

$$\varphi_{ij}(k, x_0, z_0) = \sum_{l=0}^{n_t-1} I_i(l) W_{ij}(k_t, x_0, z_0) \tag{4}$$

where $k_t = kt_s - t_{ij} - l\Delta t$, $k \in \mathbb{N}$, $t_s = 1/f_s$ is the sampling interval, $I_i(l)$ is derived from the transmembrane current source proposed in [34] and discretized at intervals of Δt for MU_i [32], $l = 0, \ldots, n_t - 1$ is the serial number of the pulse of the current source, and n_t is the total number of pulses. The weighting function W captures the effect of the

current source on the muscle fiber, and at the kth sampling time, it is calculated as

$$W_{ij}(k_t, x_0, z_0) = \frac{1}{4\pi\sigma_r\sqrt{r_f^2\frac{\sigma_z}{\sigma_r} + (av_ik_t + X_i^j - z_0)^2}} \quad (5)$$

where $a = -1$ or 1 when the current source is on the left- or right-half muscle fiber, respectively. σ_r and σ_z are the transversal and longitudinal muscle tissue conductivities, respectively, and v_i is the conduction velocity of the muscle fiber in MU_i. r_f is the distance between the observation point at $(x_0, R_m + d_r, z_0)$ and the muscle fiber. d_r is the tissue (skin and fat) thickness. As shown in Fig. 5, the distance r_j of the muscle fiber in the jth band to the reference point and the area S_{ij} of the jth band in MU_i can be obtained by arithmetic geometry.

5) MUAP Modeling: MUAP is calculated by adding the SFAPs of all the muscle fibers belonging to the same MU [35]. To reduce the computation, we propose a simplified MUAP computational method using multiband distribution of motor end-plates and the uniform distribution of muscle fibers. By doing so, we only need to calculate one SFAP in each band and then multiply the number of muscle fibers contained in this band (i.e., band area multiplied by MU fiber density). MUAP ϕ_i for MU_i is the sum of the action potentials produced by all bands, and at (x_0, z_0) and the kth step, it is calculated as

$$\phi_i(k, x_0, z_0) = \sum_{j=1}^{n_b} \varphi_{ij}(k, x_0, z_0) S_{ij}\sigma \quad (6)$$

where σ is the MU fiber density. Both skin and fat show distinctive capacitive behavior except for resistive behavior [7]. In this work, the MUAP is passed through a second-order Butterworth bandpass filter with low- and high-frequency cutoffs of 10 and 450 Hz to simulate those behaviors of skin and fat and retain effective components [36].

6) HD-sEMG Modeling: sEMG ψ observed at a point is the sum of MUAPs of all the recruited MUs with added Gaussian white noise [27], [36]. The sEMG signal observed by the electrode is equivalent to the average of the sEMG at each point across its surface [20], [26]. The HD-sEMG signal $\boldsymbol{Y} \in \mathbb{R}^{m\times n}$ observed by the electrode grid $(m+1) \times n$ at the kth step is given by $\boldsymbol{Y}(k) = [\boldsymbol{Y}_1(k) \ \cdots \ \boldsymbol{Y}_n(k)]$

$$\boldsymbol{Y}_j(k) = \left[y_{1j}(k) - y_{0j}(k) \ \cdots \ y_{mj}(k) - y_{(m-1)j}(k)\right]^T \quad (7)$$

where $y_{ij}(k)$ is the sEMG measurement at the kth step by the $(i+1)$th row and the jth column electrode, $i = 0, \ldots, m$, $j = 1, \ldots, n$, and $\boldsymbol{Y}_j$ is the jth sEMG array with m elements. Algorithm 2 further illustrates the outline of the proposed HD-sEMG model.

Algorithm 2 HD-sEMG Model-Based Computation

Input : $\boldsymbol{I}$, R_m, N, $\boldsymbol{n_f} = \{n_f^i\}$, $\boldsymbol{d_f} = \{d_f^i\}$, $\boldsymbol{v} = \{v_i\}$, σ, MFR, $\mathbf{RTE} = \{\mathrm{RTE}_i\}$, $\boldsymbol{T} = \{T_i\}$
Output: $\boldsymbol{Y}$
for $i \le N$ **do**
  Compute r_{MU}^i with n_f^i of MU_i;
  Locate MU_i using the NJA with $(0, 0)$, R_m, N, r_{MU}^i;
  Locate muscle fibers for MU_i using the NJA with $(x_{\mathrm{MU}}^i, y_{\mathrm{MU}}^i)$, r_{MU}^i, n_f^i, $\frac{d_f^i}{2}$;
  Divide MU_i into n_b bands;
  Activate MU_i with g_e and obtain the spike train;
  for $j \le n_b$ **do**
    Compute X_i^j by (1) and t_{ij} by (2) and (3);
    Update the spike trains with t_{ij};
    Compute $\varphi_{ij}(k, x_0, z_0)$ by (4) and (5);
  Compute $\phi_i(k, x_0, z_0)$ by (6);
  Update ϕ_i with a Butterworth filter and $\psi \leftarrow \psi + \phi_i$;
Update ψ with Gaussian white noise;
Compute $\{y_{ij}\}$ and $\boldsymbol{Y}$ by (7);

C. Sensor Configuration Based on the HD-sEMG Model

Table I shows the model parameters used for the sensor configuration.

TABLE I
VALUES OF THE MODEL PARAMETERS

Model Parameter (unit)	Symbol	Values
Muscle fiber length (mm)	L	80 [32]
Muscle radius (mm)	R_m	13 [32]
Muscle fiber number	N_0	30606 [28]
MU number	N	120 [31]
Recruit. threshold excitat. (% MVC)	RTE	1-80 [31]
MU fiber number	n_f	21-1720 [31]
MU fiber density (fibers/mm^2)	σ	20.79 [31]
Tissue (skin and fat) thickness (mm)	d_r	4 [32]
Muscle tissue conductivity (z) (S/m)	σ_z	0.33 [25]
Muscle tissue conductivity (xy) (S/m)	σ_r	0.063 [25]
Peak firing rate (pulses/s)	PFR	15-37.5 [31]
Minimum firing rate (pulses/s)	MFR	8 [31]
MU fiber diameter (μm)	d_f	16-75 [37]
Motor end-plate range (mm)	L_0	5 [38]
Conduction velocity (m/s)	v	2.5-5.5 [37]
Sampling frequency (Hz)	f_s	1000 [37]

1) Simulation of the Electrode Size: We simulated several commonly used sizes of square electrodes, that is, 1×1, 3×3, 4×4, 5×5, and 10×10 mm. The electrodes were placed at $(0, R_m + d_r, 0)$ in each simulation and were approximated by $n \times n$ points, $n = 3, 7, 9, 11, 21, 51$. The effect of the electrode size was assessed with the average rectified value (ARV) of the MUAP measured by the electrode [39].

2) Simulation of the Interelectrode Distance: The interelectrode distance e is the spatial sampling of the signal, and the spatial sampling frequency $x_s = v/f_s$ [22], [32], [35]. Therefore, the range of the spatial sampling frequency mainly depends on the range of conduction velocities of muscle fibers (2.5–5.5 m/s) when the sampling frequency f_s is set to the minimum value of 1000 Hz, which brings a challenge of the miniaturized design of electrodes. We simulated the interelectrode distance of 2.5, 3.5, 4.5, 5.5, 10, 15, and 20 mm,

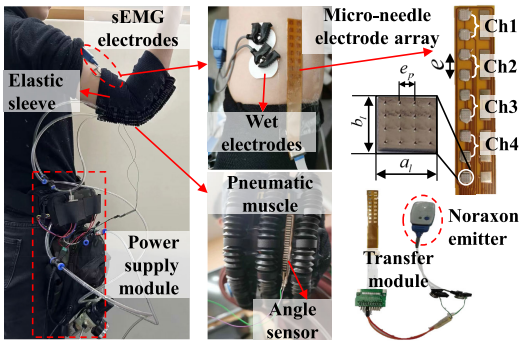

Fig. 6. Human subject experimental setup for elbow joint angle estimation.

with one electrode placed at $(0, R_m + d_r, 5)$. The effect of the interelectrode distance was assessed with the ARV, median frequency (MNF), and mean power frequency (MPF) of the MUAP [20].

The detection volume is a crucial factor for sEMG measurement, and the discussion in Section II-B1 is closely connected to the detection volume. The randomness of the MU position within the muscle cross section directly leads to different results for each simulated sEMG. Thus, referring to (6), we built an MUAP model to study the amplitude and frequency contributions of the MU to the sEMG signal when it was located at different depths along the y-axis direction. The depth away from the electrode $(0, R_m + d_r, 0)$ ranges from 5 to 11 mm with 1-mm increments.

3) Simulation of the Electrode Location: During movement, the muscles slide relative to the skin, which can cause geometrical artifacts. To avoid this, the innervation zone should be located, and the electrodes should not be placed near it. In this section, we identify the innervation zone by scanning and then determine the appropriate electrode location. We take one MU with the xy-coordinates $(0.80, 10.56)$ mm as the example to demonstrate the application. The MU radius is $r_{\mathrm{MU}} = 0.214$ mm, and the innervation zone ranges from 0.4659 to 0.4678 mm. We scan at intervals of 1 mm along the positive and negative directions of the z- and x-axis to locate the innervation zone and MU territory. The location of the innervation zone can be estimated from the differential signals because of the minimum value of ARVs along the array and the detectable phase reversal [27], [39]. The signals obtained at adjacent scan positions are differentiated, and the differential signals are normalized with the peak-to-peak value of the signal obtained at $(0, R_m + d_r, 0)$.

D. Electrode Experiment for Elbow Joint Angle Estimation

After designing the 2-D high-density microneedle electrode array, it is important to evaluate its signal acquisition performance. We conducted experiments to estimate elbow joint angle for the sEMG-based upper limb exoskeleton and verified the performance of the proposed electrode array by comparing it with commercial wet electrodes.

1) Experimental Preparation: Six healthy subjects (five males and one female; age: 24.7 ± 2.7 years; height: 168.6 ± 3.6 cm; weight: 60.3 ± 5.4 kg) with no history of neuromuscular disorders gave written informed consent before participating in the experiment. This study was approved by the Medical Ethics Committee of the School of Biomedical Engineering and Instrument Science, Zhejiang University (project identification code: 2021—39).

sEMG signals of the biceps brachii muscle were recorded simultaneously with the proposed high-density microneedle electrode array and a pair of commercial wet electrodes (22 mm in diameter and 20 mm in spacing) at a sampling frequency of 2000 Hz (Ultium, Noraxon Inc., USA). As shown in Fig. 6, they were symmetrically arranged in the muscle belly. The electrode array was connected to the emitter (Ultium, Noraxon Inc., USA) by a self-made transfer module. True angle was measured synchronously by a strain gauge angle sensor (Flex, Crownto Electronic Technology Ltd., China) on the pneumatic muscle of our self-made upper limb exoskeleton and sampled at 100 Hz. The pneumatic muscle was tightly attached to the back of the elbow with elastic sleeves.

2) Experimental Procedure: During the experiment, the upper limb exoskeleton had no power supply. The subjects were required to perform elbow flexion and extension as uniformly as possible at a speed of no more than 1 Hz under three load conditions of 0, 0.5, and 1 kg. In each condition, the subject held a dumbbell of the corresponding weight and repeated the elbow flexion and extension movement five times.

3) Data Processing: The raw sEMG signals first filtered out the low-frequency noise (baseline wandering and motion artifacts) and high-frequency background noise using a second-order Butterworth bandpass filter between 30 and 450 Hz. A second-order notch filter with a cutoff bandwidth of 4 Hz further filtered out the powerline interference at 50 Hz and its harmonics [1], [2], [40]. The angle was interpolated to meet the synchronization with sEMG. We set data windows of 200 ms in 50-ms increments, which is considered appropriate to meet real-time requirements [4], [40]. In each window, the root mean square (rms) feature of the sEMG and the mean feature of the angle were extracted [1], [2], [40]. Fivefold cross-validation was used to divide the training and testing data. The backpropagation neural network (BPNN) including three layers was used to estimate the elbow joint angle, which was widely used in previous studies [2]. The training data from each load trained one BPNN. The input to the BPNN was the rms feature of the sEMG. The number of nodes in the input and output layers depended on the number of channels of the sEMG and angle, respectively. We experimented with different numbers of nodes (ranging from 3 to 12) in the hidden layer to achieve the best results on the training data. The best structure for the BPNN with the optimal number of nodes was then obtained and used for the testing data. The sigmoid activation function was chosen in network training. The agreement between the estimated and true angles was quantitatively examined using the correlation and root-mean-squared error (RMSE).

III. Results

In this section, the HD-sEMG model is initially validated. Then, we present applications of the model in optimizing the sensor configuration. Finally, we compare the angle estimation accuracy of the proposed 2-D electrode array with that of wet electrodes.

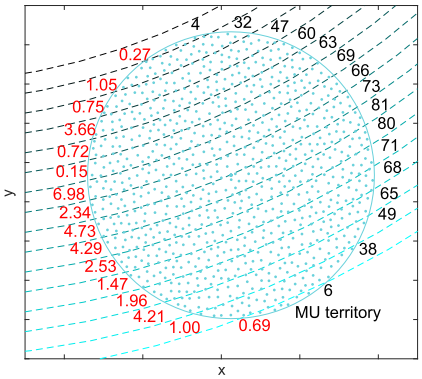

Fig. 7. Typical result of the band division and the uniform distribution of muscle fibers within one MU. The total number of muscle fibers innervated by the MU is 872. Note that the expected number of muscle fibers (band area multiplied by the MU fiber density σ) is not an integer.

TABLE II
COMPUTATION TIME WITH AND WITHOUT SIMPLIFICATION

	Case 1 (s)	Case 2 (s)	NRMSE (%)
MUAP	30.3 ± 15.8	123.3 ± 9.7	0.48 ± 0.41
sEMG	136.3 ± 11.5	3929.5 ± 544.2	3.37 ± 0.07

A. Model Validation and Performance Demonstration

Ensuring the uniformity of muscle fibers is one of the requirements for the simplified computational method of MUAP generation. Fig. 7 shows a typical result of band division and uniform distribution of muscle fibers within one MU. The black numbers represent the actual number of muscle fibers generated in each band, and the red numbers represent the absolute difference between the expected number and the actual number of muscle fibers in each band. The average value of these red numbers is 2.3, with a maximum of 6.98. The maximum value accounts for 9.56% of the expected number of the corresponding band. The fiber density of each band is calculated, and the actual MU fiber density is averaged across all the bands in the MU, with a value of 21.08 ± 1.22 fibers/mm^2 ($\sigma = 20.79$ fibers/mm^2).

To evaluate the computational cost, we simulated the MUAP of one MU (see Fig. 7) and a single-channel sEMG with (case 1) and without (case 2) the proposed simplified MUAP computational method. Each case was executed three times, and a representative example of the simulated sEMG in case 1 is shown in Fig. S1 in the supplementary material. Table II lists the average computation time. We also computed the normalized RMSE (NRMSE) between the signals generated in the two cases to ensure consistency between the signals. The NRMSE of the MUAP was averaged across all the MUs. With the proposed method, the errors are 0.5% and 3.5% on the MUAPs and the sEMG, respectively. The computation times are reduced by 75.4% and 96.5% for the MUAP and the sEMG, respectively. The computation time is greatly reduced without sacrificing the simulation accuracy, with the average computation time of a single-channel sEMG being less than 3 min. Furthermore, the representative computation time for HD-sEMG on an 8×8 electrode grid simulated with the proposed computational method is less than 10 min.

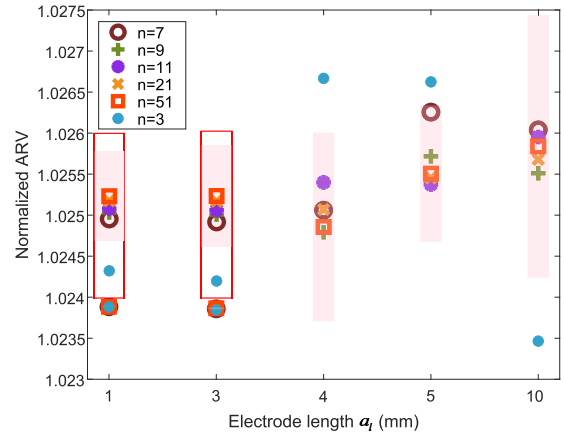

Fig. 8. Normalized ARV of one MU from electrodes with different sizes. The calculated ARV was normalized by the value calculated at the electrode center. The pink-filled area represents the 90% confidence interval. Shown in the two red boxes are enlarged images of the results for electrodes of size of 1 and 9 mm^2.

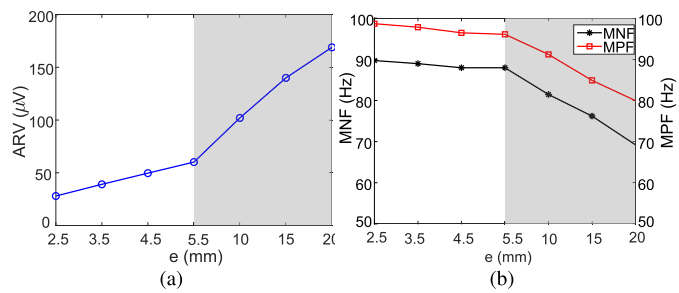

Fig. 9. (a) ARV and (b) MNF and MPF values as a function of interelectrode distance. The interelectrode distance is taken from 2.5 to 5.5 mm with 1-mm increments. The shaded area shows the results of the commonly used interelectrode distance.

B. Sensor Configuration Based on the HD-sEMG Model

1) Effects of the Electrode Size: Fig. 8 shows the ARV values that were normalized by the value at the electrode center. A larger n results in a more accurate ARV value; therefore, the ARV when $n = 51$ is taken as the estimate of the actual ARV of the electrode. We then calculated the 90% confidence interval of the actual ARV, and the n corresponding to the ARV outside the confidence interval is regarded as a false approximation to the electrode. The calculation concludes that the cases of $n \geq 9$ approximate the electrodes well. Correspondingly, the distance between points does not exceed 1.25 mm, and this guides the design of the pin-to-pin distance e_p of the microneedle electrode.

2) Effects of the Interelectrode Distance: Fig. 9(a) shows the computed ARV. Fig. 9(b) shows the MNF, and MPF values over varying interelectrode distances. The ARV values increase with larger interelectrode distances, while the MNF and MPF values decrease.

Fig. 10(a) shows the MUAP aptitude profiles in the time. Fig. 10(b) shows the frequency domains under different MU depths. When the MU is located deeper in the muscle, the generated MUAP has a smaller amplitude. In Fig. 10(b), with increased depths, the MUAP signal's high-frequency components gradually disappear, and the amplitude of each frequency component reduces as well.

TABLE III
PERFORMANCE COMPARISON (MEAN ± SD) OF DIFFERENT METHODS UNDER THREE LOADS

Metrics	Load (kg)	Proposed						Wet	Improvement (%)
		Ch1	Ch2	Ch3	Ch4	Chs	All		
RMSE (°)	0	25.63 ± 9.60	23.22 ± 10.17	23.22 ± 11.80	22.77 ± 9.90	23.71 ± 10.31	17.07 ± 5.26	22.68 ± 8.81	**22.35**
	0.5	14.03 ± 5.46	13.80 ± 5.34	14.06 ± 5.93	14.07 ± 6.06	13.99 ± 5.63	11.00 ± 4.20	13.59 ± 4.80	**19.11**
	1.0	18.39 ± 5.31	18.19 ± 6.70	17.78 ± 4.60	16.29 ± 4.45	17.66 ± 5.33	14.06 ± 5.09	17.18 ± 6.11	**18.03**
Correlation	0	0.70 ± 0.19	0.74 ± 0.17	0.73 ± 0.19	0.73 ± 0.21	0.73 ± 0.19	0.85 ± 0.09	0.75 ± 0.15	**14.74**
	0.5	0.89 ± 0.10	0.89 ± 0.10	0.89 ± 0.10	0.88 ± 0.13	0.89 ± 0.11	0.94 ± 0.04	0.91 ± 0.06	**3.33**
	1.0	0.83 ± 0.11	0.82 ± 0.12	0.84 ± 0.10	0.86 ± 0.09	0.83 ± 0.11	0.88 ± 0.11	0.84 ± 0.12	**4.63**

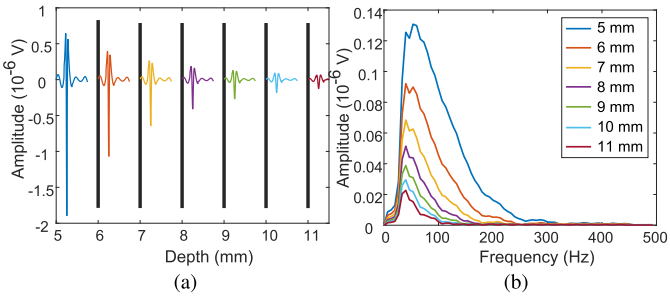

Fig. 10. (a) Time- and (b) frequency-domain images of MUAPs with the MU located at different subcutaneous depths.

3) Electrode Location: Fig. 11(a) shows the normalized ARV values along the z- and x-axis directions, respectively. The minimum values of ARVs occur at $z = 1$ mm and $x = 0.25$ to 1.25 mm. Fig. 11(b) and (c) show the waveforms of MUAP along the z- and x-axis directions, respectively. The ranges between the two planes are where the innervation zone and the MU territory are located. Note that due to the small extent of the innervation zone, the two blue planes cannot be seen unless the image is enlarged. The potentials with phase reversal occur at $z = 0.75$ and 1 mm and $x = 0$ and 1.5 mm. Therefore, the position of the innervation zone is in the range of $(0.375, 0.5)$ mm, the position of the MU territory is in the range of $(0, 1.5)$ mm, and the position of the MU territory center is in the range of $(0.214, 1.286)$ mm. The observed results are consistent with the locations of the innervation zone and MU territory. Scan positions are indicated by the black dots spaced 1 mm apart. To locate the positions of the innervation zone and MU territory precisely, the scan positions can be further refined with increased spatial resolution, as the red dots shown in the plots. Moving away from the innervation zone, the ARV value first increases, then decreases, and finally remains stable. In the direction away from the MU territory, the ARV value first becomes larger and then smaller. The difference between adjacent ARVs reduces. The position of the maximum value of ARV is the optimal electrode location, and therefore, the signal-to-noise ratio is high.

C. Electrode Experiment for Elbow Joint Angle Estimation

Based on the simulation results of electrodes in Section III-B and the detailed discussion in Section IV, the size of a single microneedle electrode is $a_l = b_l = 2.5$ mm, and there are 4×4 microneedles with a height of 230 μm and a spacing of $e_p = 0.625$ mm. The interelectrode distance is $e = 6$ mm. As shown in Fig. 6, to achieve consistent coverage with the wet electrode, the high-density microneedle electrode array introduced four differential channels along the muscle direction, denoted as Ch1, Ch2, Ch3, and Ch4 from proximal to distal. Five different BPNNs were modeled for each channel (Ch1, Ch2, Ch3, Ch4, and Wet). To further evaluate the performance of electrode space utilization, another BPNN was modeled using the data of all the channels of the high-density microneedle electrode array (All).

The overall estimation results of the single-channel and all-channel methods under three loads are shown in Fig. 12 and Table III. The correlation and RMSE were averaged across all subjects and trials. Chs represents the average single-channel estimation result of the high-density microneedle electrode array. Wilcoxon rank-sum test was performed on two factors: the method and the load. There was no significant difference between the single-channel estimation of the high-density microneedle electrode array ($p > 0.05$). The single-channel estimation of the high-density microneedle electrode array was comparable to that of the wet electrode under any load ($p > 0.05$), but lower than the all-channel estimation of the high-density microneedle electrode array ($p < 0.05$). The all-channel method of the high-density microneedle electrode array achieved the smallest RMSEs and the highest correlations under 0.5 kg load ($p < 0.05$). The load variation had an essential influence on the performance of angle estimation. The 0.5-kg load showed a better correlation than the 0-kg load for the all-channel high-density microneedle electrode array and the wet electrode ($p < 0.05$). As shown in Table III, the average improvement in the correlation of the high-density microneedle electrode array with high space utilization (all-method) under three loads is 14.74%, 3.33%, and 4.63%, and the average improvement in the RMSE is 22.35%, 19.11%, and 18.03%. On average, the correlation and RMSE are 7.56% and 19.83% better than those of the wet electrode, respectively.

The estimation results of the single-channel and all-channel methods under three loads for each subject are shown in Fig. 13 and Tables IV and V. The correlation and RMSE were averaged across trials. The load variation had a more significant influence on the performance of angle estimation for individual subjects ($p < 0.01$), such as subject S1. The RMSE comparison of the single-channel method of the high-density microneedle electrode array with the all-channel method under three loads for subject S6 showed that the performance between channels varied with the load (see Fig. 13). Fig. 14 shows the estimated angles using each of the six methods under 0.5-kg load for subject S2 compared with the measured values. The correlation is 0.96, 0.98, 0.87, 0.95, 0.98, and 0.93, and the RMSE is 12.42°, 8.25°, 22.34°, 14.92°,

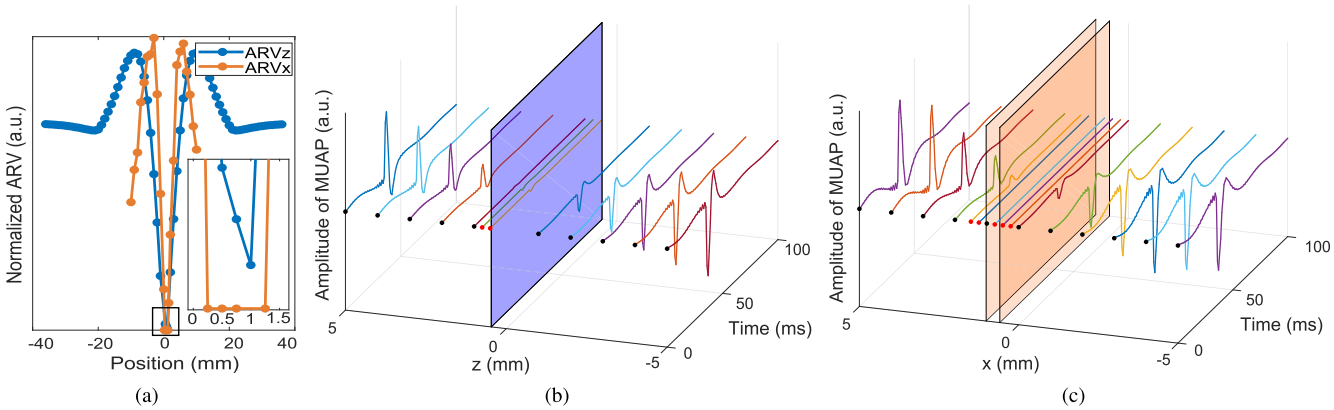

Fig. 11. Normalized ARVs and scan images of differential signals in the z- and x-axis directions. The scan ranges in the z- and x-axis directions are $[-36, 36]$ and $[-10, 10]$ mm, respectively. (a) Minimum values of ARVs occur at $z = 1$ mm and $x = 0.25$ to 1.25 mm. (b) Potentials with phase reversal occur at $z = 0.75$ and 1 mm. (c) Potentials with phase reversal occur at $x = 0$ and 1.5 mm.

TABLE IV

COMPARISON OF RMSE (°, MEAN ± SD) FROM DIFFERENT METHODS UNDER THREE LOADS FOR ALL SUBJECTS

Subject	Load (kg)	Proposed						Wet	Improvement (%)
		Ch1	Ch2	Ch3	Ch4	Chs	All		
S1	0	30.25 ± 11.72	22.88 ± 9.17	23.32 ± 9.81	28.98 ± 8.94	26.36 ± 9.75	17.71 ± 2.45	22.76 ± 9.45	22.19
	0.5	10.43 ± 1.86	10.80 ± 2.51	10.33 ± 1.24	10.58 ± 2.10	10.53 ± 1.83	8.46 ± 1.16	10.61 ± 2.92	20.26
	1.0	14.24 ± 1.64	13.34 ± 2.07	13.54 ± 2.66	14.24 ± 3.93	13.84 ± 2.53	12.24 ± 3.21	11.54 ± 1.02	−6.07
S2	0	31.20 ± 10.90	28.54 ± 14.68	30.92 ± 17.22	23.06 ± 14.19	28.43 ± 13.65	20.65 ± 6.34	33.23 ± 8.40	37.86
	0.5	19.21 ± 2.95	17.55 ± 5.56	17.42 ± 6.63	15.49 ± 5.50	17.42 ± 5.08	16.02 ± 4.93	20.26 ± 3.18	20.93
	1.0	17.20 ± 4.16	12.66 ± 6.26	20.28 ± 3.83	16.21 ± 4.86	16.59 ± 5.27	12.41 ± 5.01	18.14 ± 3.84	31.59
S3	0	19.29 ± 5.49	20.64 ± 11.99	20.47 ± 0.11	16.74 ± 10.57	19.28 ± 6.60	13.07 ± 6.29	20.07 ± 4.53	34.88
	0.5	17.30 ± 3.50	17.19 ± 1.54	21.44 ± 3.26	19.87 ± 4.25	18.95 ± 3.37	14.35 ± 1.89	16.22 ± 4.06	11.53
	1.0	19.96 ± 5.95	16.83 ± 15.74	15.26 ± 3.73	12.18 ± 4.16	16.06 ± 7.34	14.55 ± 2.69	20.17 ± 8.34	27.86
S4	0	23.83 ± 7.47	17.84 ± 5.95	17.00 ± 13.09	18.36 ± 5.19	19.26 ± 8.10	16.83 ± 5.67	20.76 ± 3.03	18.93
	0.5	10.18 ± 2.66	10.91 ± 1.74	13.05 ± 1.91	12.05 ± 1.84	11.55 ± 2.20	9.41 ± 2.24	10.66 ± 3.62	11.73
	1.0	18.30 ± 4.77	22.50 ± 5.48	16.02 ± 4.71	17.32 ± 2.77	18.54 ± 4.86	14.11 ± 4.03	16.96 ± 7.38	16.80
S5	0	14.94 ± 2.23	15.85 ± 1.98	14.93 ± 7.02	15.58 ± 5.08	15.32 ± 4.23	13.98 ± 3.06	13.15 ± 2.03	−6.31
	0.5	20.36 ± 1.55	19.28 ± 5.37	16.69 ± 7.62	20.67 ± 7.26	19.25 ± 5.70	12.61 ± 2.36	14.02 ± 4.00	10.06
	1.0	23.82 ± 5.58	22.77 ± 5.03	22.59 ± 4.04	21.37 ± 3.40	22.64 ± 4.31	19.54 ± 6.89	21.43 ± 6.68	8.82
S6	0	30.08 ± 3.01	30.97 ± 8.37	29.80 ± 6.59	29.39 ± 7.23	30.06 ± 6.09	17.74 ± 6.80	24.16 ± 6.84	26.57
	0.5	8.03 ± 1.35	8.44 ± 0.75	8.38 ± 1.39	8.05 ± 0.53	8.22 ± 1.00	6.48 ± 1.00	10.83 ± 2.66	40.17
	1.0	17.75 ± 6.22	20.23 ± 2.88	17.46 ± 2.65	13.95 ± 3.44	17.35 ± 4.37	11.80 ± 3.95	16.66 ± 6.47	29.17

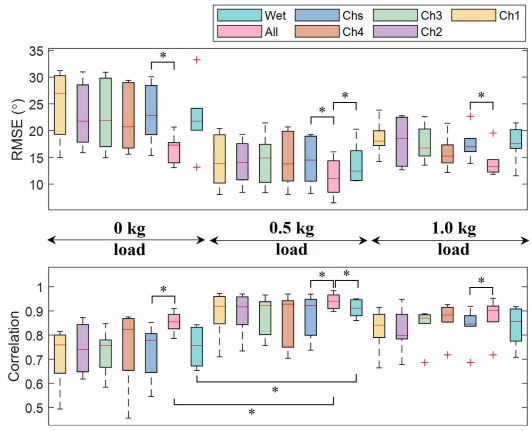

Fig. 12. Overall correlation and RMSE results of the single-channel and all-channel methods under three loads. $*$, $p < 0.05$.

8.66°, and 16.37° from Fig. 14(a)–(f). The estimations are better than those of the wet electrode except for Ch3. The all-channel method estimates smoother angles. Overall, the effectiveness of the high-density microneedle electrode array and its configuration was verified by experiments.

IV. DISCUSSION

In addition to various complex properties such as fiber orientation, location, size, and anisotropy of tissues, the muscle fiber number also hinders further exploration of sEMG. Since one MU may be composed of hundreds or thousands of muscle fibers, the simulation costs of MUAP, sEMG, and HD-sEMG are high. In our method, the SFAP only needs to be calculated once for each MU band instead of calculating SFAPs at each time for all the muscle fibers. As shown in Table II, the computation time of the MUAP with and without simplification is 30.3 ± 15.8 and 123.3 ± 9.7 s, and the computation time of the sEMG is 136.3 ± 11.5 and 3929.5 ± 544.2 s, respectively. The simplified computational method reduces the computation time of an 8×8 HD-sEMG from more than 1 h to 10 min and achieves a sixfold reduction compared with the state-of-the-art. Therefore, compared with the traditional model without simplification, the proposed model greatly improves

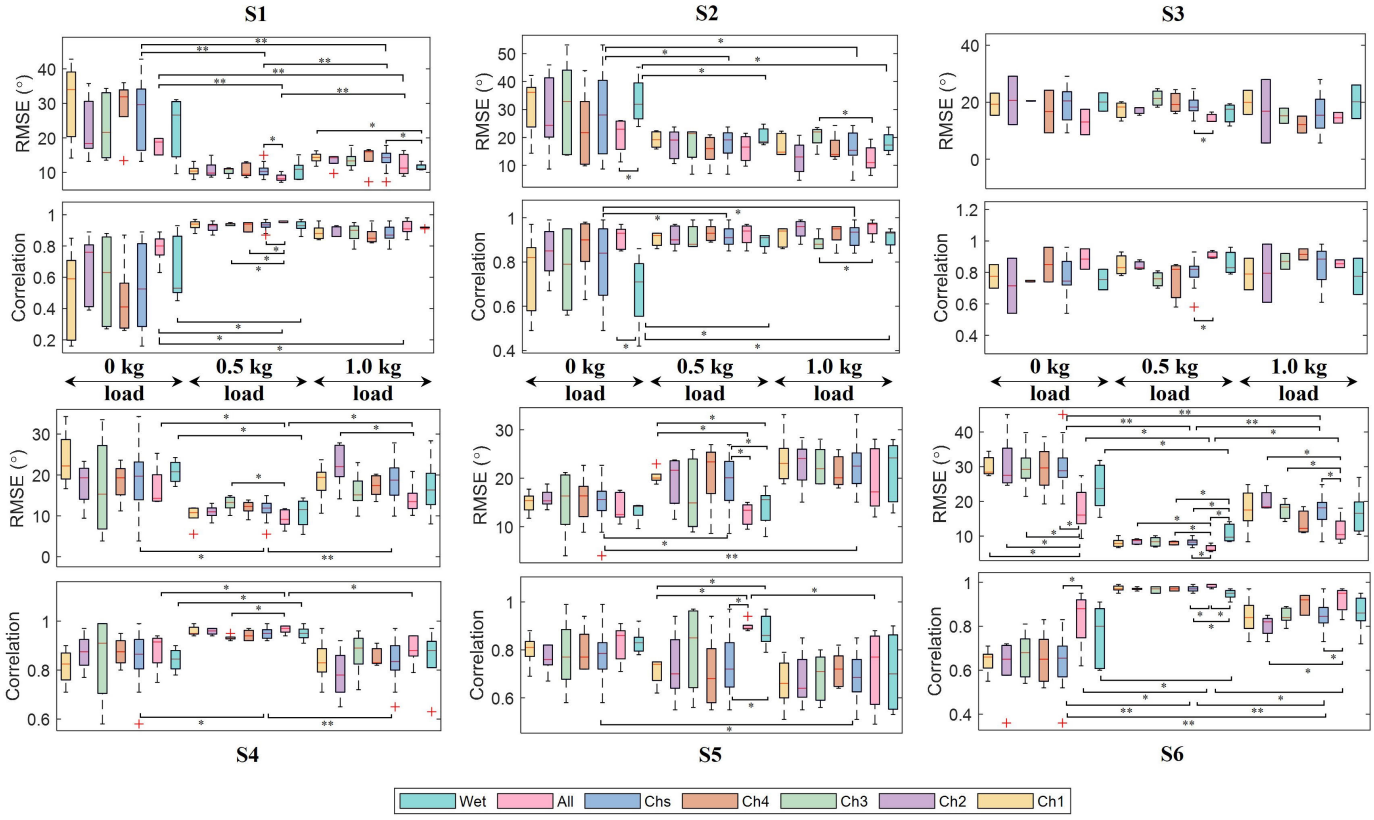

Fig. 13. Correlation and RMSE results of the single-channel and all-channel methods under three loads for each subject. $*$, $p < 0.05$. $**$, $p < 0.01$.

TABLE V

COMPARISON OF CORRELATION (MEAN ± SD) FROM DIFFERENT METHODS UNDER THREE LOADS FOR ALL SUBJECTS

Subject	Load (kg)	Proposed						Wet	Improvement (%)
		Ch1	Ch2	Ch3	Ch4	Chs	All		
S1	0	0.49 ± 0.30	0.65 ± 0.23	0.58 ± 0.29	0.46 ± 0.25	0.55 ± 0.26	0.79 ± 0.10	0.65 ± 0.22	21.54
	0.5	0.93 ± 0.03	0.92 ± 0.03	0.94 ± 0.01	0.92 ± 0.03	0.93 ± 0.03	0.96 ± 0.01	0.93 ± 0.04	3.23
	1.0	0.89 ± 0.05	0.89 ± 0.04	0.88 ± 0.07	0.87 ± 0.06	0.88 ± 0.05	0.92 ± 0.05	0.92 ± 0.00	0
S2	0	0.74 ± 0.19	0.84 ± 0.12	0.77 ± 0.19	0.87 ± 0.14	0.81 ± 0.16	0.91 ± 0.05	0.67 ± 0.17	35.82
	0.5	0.90 ± 0.03	0.91 ± 0.05	0.91 ± 0.06	0.93 ± 0.04	0.92 ± 0.04	0.92 ± 0.05	0.89 ± 0.03	3.37
	1.0	0.91 ± 0.05	0.95 ± 0.04	0.89 ± 0.04	0.93 ± 0.05	0.92 ± 0.05	0.95 ± 0.04	0.91 ± 0.04	4.40
S3	0	0.78 ± 0.11	0.72 ± 0.25	0.75 ± 0.01	0.85 ± 0.16	0.77 ± 0.13	0.89 ± 0.09	0.76 ± 0.09	17.11
	0.5	0.85 ± 0.08	0.84 ± 0.03	0.76 ± 0.06	0.75 ± 0.15	0.80 ± 0.09	0.91 ± 0.03	0.86 ± 0.09	5.81
	1.0	0.79 ± 0.14	0.80 ± 0.26	0.87 ± 0.07	0.92 ± 0.05	0.84 ± 0.13	0.86 ± 0.04	0.78 ± 0.16	10.26
S4	0	0.82 ± 0.08	0.87 ± 0.08	0.85 ± 0.19	0.88 ± 0.06	0.85 ± 0.11	0.88 ± 0.09	0.84 ± 0.05	4.76
	0.5	0.96 ± 0.02	0.96 ± 0.01	0.93 ± 0.01	0.94 ± 0.02	0.95 ± 0.02	0.97 ± 0.02	0.95 ± 0.03	2.11
	1.0	0.84 ± 0.09	0.78 ± 0.10	0.87 ± 0.09	0.85 ± 0.04	0.84 ± 0.09	0.89 ± 0.06	0.85 ± 0.13	4.71
S5	0	0.80 ± 0.07	0.77 ± 0.06	0.78 ± 0.15	0.80 ± 0.09	0.79 ± 0.09	0.83 ± 0.08	0.83 ± 0.05	0
	0.5	0.71 ± 0.06	0.73 ± 0.15	0.80 ± 0.18	0.70 ± 0.16	0.74 ± 0.14	0.90 ± 0.02	0.88 ± 0.07	2.27
	1.0	0.66 ± 0.11	0.68 ± 0.12	0.69 ± 0.10	0.72 ± 0.08	0.69 ± 0.10	0.72 ± 0.17	0.71 ± 0.17	1.41
S6	0	0.64 ± 0.06	0.62 ± 0.15	0.67 ± 0.11	0.65 ± 0.12	0.65 ± 0.11	0.83 ± 0.13	0.76 ± 0.15	9.21
	0.5	0.97 ± 0.01	0.97 ± 0.01	0.97 ± 0.02	0.97 ± 0.01	0.97 ± 0.01	0.98 ± 0.01	0.95 ± 0.02	3.16
	1.0	0.84 ± 0.09	0.80 ± 0.05	0.85 ± 0.04	0.90 ± 0.05	0.85 ± 0.06	0.92 ± 0.06	0.86 ± 0.09	6.98

the simulation efficiency, enabling a large number of muscle fibers to be simulated with very low computational costs.

Based on the fact that MU territories and muscle fibers are uniformly distributed, we developed the NJA algorithm for the generation of uniformly distributed points. The shapes of the muscle and MU territories are assumed to be circular, and this algorithm can be extended to a variety of physiological shapes. As shown in Fig. 7, the NJA algorithm creates highly accurate uniform distribution. The quantitative analysis of the muscle fiber number in each band shows that the NJA algorithm is able to generate the desired number of muscle fibers according to the area and the desired fiber density, which also enables the proposed simplified MUAP computational method.

Fig. 8 demonstrates the ARVs of the same MU measured by electrodes of different sizes. The normalized ARV values measured by electrodes with a size not greater than 3 × 3 mm

are approximately equal to 1, which presents that a small-size electrode can be regarded as a point. As the electrode size increases, the normalized ARV values deviate from 1, suggesting that the potentials in the interval where the distance difference between the electrode and the MU is small are not much different. This observation allows us to realize the simplified model of sEMG, such as the method of the macroscale MU electrical source mentioned in [28] and the notion of the isopotential layer borrowed in this study and [20], to save the computational cost and speed up the calculation.

The average of the nonpoint electrode potential (see the transfer function in [22]) and the interelectrode distance (see Fig. 9) brings a low-pass filtering effect. Therefore, the larger the electrode size and interelectrode distance, the better the noise suppression and the higher the signal-to-noise ratio, but the spatial sampling frequency is also reduced. On the contrary, the smaller the size, the lower the signal-to-noise ratio, and the more easily the signal is distorted. Therefore, it is recommended to design microneedle electrodes to expand the area of small electrodes without affecting the spatial sampling frequency. This design not only increases the signal-to-noise ratio but also overcomes the insulation problem of the skin stratum corneum [22]. Small microneedle electrodes with sizes $a_l, b_l \leq 3$ mm, pin-to-pin distances $e_p \leq 1.25$ mm, and interelectrode distances $e \leq 6$ mm are recommended, as shown in Figs. 8 and 9. The conclusion of electrode parameters is similar to that of the studies in [22] and [23]. From this viewpoint, most of the sEMG electrodes commonly used do not meet such a design [23].

Fig. 10 simulates the signal characteristics when the MU is at different subcutaneous depths, revealing differences in amplitude and frequency content. The detection volume is mainly derived from SFAPs within 10 mm from the electrode, which is in a similar range to the detection depth estimated by previous analytical models in [23] and [27]. The measured signal is only the active signal of the muscle surface, so it is called the sEMG. The potentials generated during source conduction along the muscle fiber and extinction at the tendon diffuse across the skin surface. The signals measured by electrodes may be mixed with contributions from sources of other muscles, a phenomenon known as crosstalk. Although crosstalk is unavoidable, to ensure the purity of the target signal, it is necessary to reduce the interfering signal as much as possible. The simple approach is to place the electrodes in the middle of the muscle belly away from the innervation zone. In addition, the border of the HD-sEMG sensor should be no less than 10 mm away from the muscle border, which defines its maximum limit of size.

There are still limitations to be overcome in the future. Additional visualization techniques, such as ultrasound and diffusion tensor imaging, are necessary to improve the anatomical and physiological accuracy of the model to present a subject-specific model. Furthermore, obtaining simulations of the HD-sEMG during dynamic contraction is attractive for various applications.

In Section III-C, we designed a high-density microneedle electrode array based on simulation results for sEMG measurement of the upper limb exoskeleton. Although the advantages

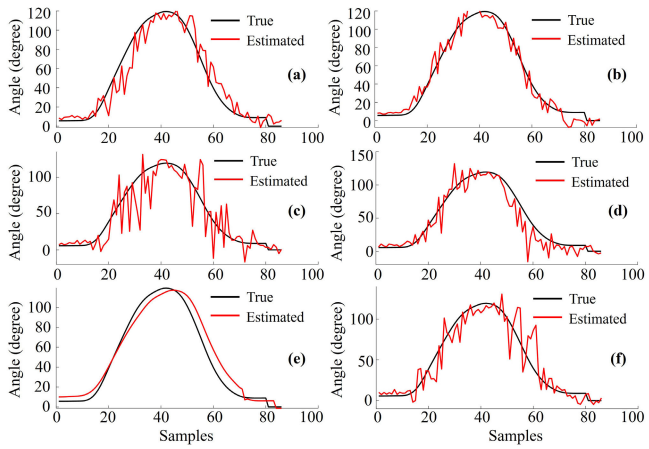

Fig. 14. Comparison of the estimated elbow joint angle (red) with the measured angle (black) under 0.5 kg load for subject S2. (a) Ch1. (b) Ch2. (c) Ch3. (d) Ch4. (e) All. (f) Wet.

of microneedle electrodes for long-term sEMG monitoring without skin preparation are popular in the literature [5], [7], this work focuses on high density, high spatial resolution, and high space utilization. As shown in Fig. 13 and Table IV, the mean RMSEs of the single-channel (Ch4) and all-channel (All) methods of the high-density microneedle electrode array under the 0.5 kg-load for subject S6 are 8.05° and 6.48°, respectively, while the mean RMSE of the wet electrode is 10.83°. The experimental results show that the rigorously configured microneedle electrode can match or even outperform the wet electrode, especially the multichannel array ($p < 0.05$). However, the performance of high-density electrode array is susceptible to interference from bad channels, such as Ch3 in Fig. 14. The estimation accuracy can be maximized by screening the channels (combination and removal), and it will be significantly better than the traditional single-channel estimation. We also observe that the estimation performance with load is superior to that without load ($p < 0.05$) [41]. Moreover, the dominant channel in the high-density electrode array varies under different loads. These are mainly due to different activation patterns of muscles, with different numbers and positions of MUs being recruited and discharged.

The high-density electrode array captures spatial information and thus has greater potential for practical and clinical applications. The performance of the neural-driven approach is superior to the traditional sEMG-driven approach, especially during fast movements [40]. In the future, we will attempt to decode MUAPs from HD-sEMG recorded by the proposed high-density microneedle electrode array for more robust and precise control of exoskeleton robots.

V. Conclusion

Understanding muscle composition, anatomical and physiological properties, and the process of sEMG generation is critical for extracting accurate motion and intention information from sEMG signals. This work presented modifications to the locations of MUs and muscle fibers and the geometry of motor end-plates for HD-sEMG recordings. Based on the modifications, a simplified computational method was

proposed to improve the simulation efficiency. The 2-D high-density microneedle electrode configuration was carried out using the neuromuscular analytical model for HD-sEMG. The application of the configured microneedle electrode array in elbow joint angle estimation was demonstrated for verification and comparison. The model can be used as an effective tool for the development and analysis of HD-sEMG signals, and the developed 2-D high-density microneedle electrode array shows promising potential for practical applications.

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